Rethinking the Implementation of Artificial Intelligence for Sustainable Education in Africa: Challenges and Solutions

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RETHINKING THE IMPLEMENTATION OF ARTIFICIAL INTELLIGENCE FOR A SUSTAINABLE EDUCATION IN AFRICA: CHALLENGES AND SOLUTIONS

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ABSTRACT

Recent socio-economic trends have made Artificial intelligence (AI) a vital institutional force driving development among countries with optimal opportunities and costs. Among researchers in this domain, the benefit of AI is highly espoused, having been underexplored in Africa. However, the outbreak of the COVID-19 pandemic has highlighted the need to strengthen the education sector, given that many schools have migrated their teaching and learning online. While face-to-face teaching was the norm, the transition to online teaching has brought about the need to rethink the use of Information Technology to strengthen teaching and learning. To proffer solutions for the implementation of AI in Africa, there is the need to understand the challenges. Therefore, in this chapter, we explore the possible challenges that hinder the implementation of AI in Africa. Further, we propose solutions for the implementation of AI in the education sector, especially in this era of the COVID-19 pandemic. The solutions stem from rethinking the role of AI in the education sector. Finally, a conceptual framework that synthesises the problems and the proposed solution is developed. We envision that the proffered solutions can mitigate the deepening misconceptions and challenges bedevilling AI implementation in Africa.

Keywords: Artificial intelligence; education; sustainable education; technology; Africa; design science

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3.1 INTRODUCTION

Education is vital for the growth and human capital development of any nation. Undoubtedly, the influence and impact of the COVID-19 pandemic have been drastic on education (Kandri, 2020). Historically, COVID-19 has been reported as causing the largest disruption in the global educational system (Mogaji, 2020; Pokhrel & Chhetri, 2021). Africa is equally facing the negative effects. For instance, teaching and learning timelines and trajectories have been distorted, with governments and institutions having little or no time to prepare for the new academic calendar (Daniel, 2020; Mogaji, Jain, Maringe, & Hinson, 2022). It is estimated that about 250 million children, at both the secondary and primary levels, were out of school due to the closure caused by the COVID-19 pandemic (Cummins, 2020). As learning losses are associated with school closures (Kaffenberger, 2021), the damage caused by the pandemic on the academic growth of children in Africa is likely to be severe. Face-to-face learning was seriously hampered due to the restrictions on movement and the enforcement of social distancing (Pokhrel & Chhetri, 2021). This is dire with Africa, where many countries are used to face-to-face teaching and learning. Teachers had the shocking realisation of the steep difference in preparing, delivering and assessing students, with little support or training from their institutions (de Jong, 2020).

Similarly, even higher educational institutions, like the universities, globally were practically pushed into remote or online learning (Gallagher & Palmer, 2020). As a result, many universities quickly transferred their courses to online platforms (Aristovnik, Keržič, Ravšelj, Tomaževič, & Umek, 2020). This has led to attempts to radically revolutionise universities' mode of operation by rethinking the traditional higher education business model to a digitisation-focused one and moving beyond remote learning strategies (Gallagher & Palmer, 2020).

According to Pandey, Pal and others (2020), most schools, colleges and universities worldwide have moved their classes online through video conferencing platforms, such as Zoom and Google Meet, since the beginning of the lockdown era. At the core of the transformation agenda of educational institutions in the new normal is the role of digital transformation technologies. These technologies, such as Cloud Computing, Internet-of-Things, Blockchain and Artificial Intelligence (AI), represent a great majority of the digital technologies being adopted by organisations as part of their transformation drive (Pandey et al., 2020). While the Global North is poised to leverage information and communication technology (ICT), especially AI, to augment teaching and learning activities, Africa is challenged to leverage AI to improve the education sector. This conceptual chapter aims to explore and provide guidance on the implementation of AI in the education sector in Africa. To address this aim, two research questions are proposed: (1) *What are the challenges of AI implementation in the education sector in Africa, and* (2) *What solutions could address the identified challenges?* Literature on AI and education from various sources was analysed, and the outcome led to understanding the challenges in AI implementation in Africa. Based on the identified challenges, solutions have been proposed for AI implementation towards the fourth industrial revolution (4IR) in Africa.

The rest of the chapter covers the background of the study, which delves into AI and its need in the education sector. This follows with Africa's existing AI implementation challenges, reflecting on three categories of AI implementation challenges in Africa. The next section delves into the solutions to the challenges by rethinking the role of AI for sustainable education in Africa. Finally, the chapter concludes by summarising the key findings and setting an agenda for future research.

3.2 AI IN EDUCATION

Indeed, information technology (IT) has brought a massive improvement in disseminating knowledge through the realisation of online learning and other similar pedagogy. Many African universities resumed academic activities via e-learning and videoconferencing platforms (Aristovnik et al., 2020). Migrating academic work online or to virtual environments requires that schools have the needed IT infrastructure to support teaching and learning processes. In addition, since both teachers and students were expected to work from home, they were also required to own digital devices that could support online learning and be well trained in using such devices, and the requisite software such as e-learning systems, video editing software and videoconferencing tools. Some experts posit teachers need weeks of training to adequately prepare and perform their teaching duties online (Adeyanju, Ajilore, Ogunlalu, Onatunji, & Mogaji, 2022). Already, UNESCO's International Task Force (2021) lament that Sub-Saharan African (SSA) countries are facing a crisis in the sufficiency of qualified teachers to serve the needs of the ever-growing demand for schools. Coupled with this is the inadequate training given to many teachers across the sub-region (Olaleye, Ukpabi, & Mogaji, 2021). In addition, others expressed fear that the disruption of academic activities due to the interruption of school programmes and the absence of remote educational technologies will cause delays in the ability to complete school programmes (Umviligihozo et al., 2020).

It is reported that the online learning experiences shared by teachers and students in Africa could be described simply as 'challenge-ridden online learning' (Adarkwah, 2021). Where efforts to online learning were present, issues regarding lack of adequate learning resources, difficulties in self-engaged learning, and absence of effective supervision and parental support in home learning hampered the process (Biwer et al., 2021; Owusu-Fordjour, Koomson, & Hanson, 2020). This shows the lack of requisite IT infrastructure and training of teachers and students towards applying virtual or online learning pedagogy. In addition, students generally face difficulty adapting to the relatively new approach to teaching and learning (Biwer et al., 2021). Thus, it is reported that students could not undertake any effective teaching and learning activities from home due to the very ineffective online learning system (Owusu-Fordjour et al., 2020). The pandemic has exposed the inequality resulting from the digital divide in the educational system (Zhong, 2020) prevalent in the sub-region. The effect of this is clear in work carried out in Burkina Faso, Ethiopia and Nigeria, which showed that adolescents, predominantly in primary and secondary schools, had their schooling disrupted during the peak of the pandemic (Wang et al., 2021). Many of these adolescents had no access to any remote, virtual or online learning experience, as most schools were closed to any form of teaching and learning. Mobile devices could be strategic in helping to address the digital gap in assessing teaching and learning during disruptions in many countries, as they have been widely accessible and affordable (Kizilcec, Chen, Jasińska, Madaio, & Ogan, 2021; Mogaji, Olaleye, & Ukpabi, 2020).

Nevertheless, countries like Ghana prohibit mobile devices in high schools among students (Kolog, Tweneboah, Devine, & Adusei, 2018). The consequence is that such students could not leverage such technologies to continue academic work. This was also due to the non-existence of the needed online/virtual learning systems or would they have trained their students and teachers in their use. For this reason, many second-cycle schools had to remain closed with no academic activity taking place, even online (Winthrop, 2020).

Following the relaxed restriction on movement and strict enforcement of social distancing, many schools resume academic work by combining face-to-face and online learning. While schools in the urban areas could harness the capabilities of these technology-based pedagogies and therefore continue learning, those in the rural areas could not due to lack of accessibility. These besetting accessibility challenges were mainly due to lack of internet, stable electricity, poor or unavailable IT infrastructure, lack of technical know-how and inability to own digital devices that support online learning (Owusu-Fordjour et al., 2020). Despite the challenges, the pandemic has spurred the application of novel approaches to daily lives, especially with technological inventions and innovations in the educational sector. Due to the pandemic, educational institutions show diversification and massive adoption of online learning approaches in Africa (Adarkwah, 2021; Sharma, Soetan, Farinloye, & Noite, 2022).

Recent socio-economic trends have made AI a vital institutional force driving development in countries with optimal opportunities and costs. Yet, among several researchers in this domain, AI's benefit is highly espoused and has been under-explored. Some scholars suggest that AI's benefits are only imaginary (Basri, 2020), although the opportunities brought by AI in advanced countries are immense. Yet, the concept seems to be in its infancy in developing nations, especially on the African continent. Moreover, the concept is elusive in its practical sense, as there are several challenges relative to the continent's social, technical, or infrastructure.

Nonetheless, the rate of the digital divide as exacerbated during the COVID-19 pandemic exposed the disparity among developing countries (Kolog, Egala, Amponsah, Devine, & Sutinen, 2021). The technology that allows non-human machines to have intelligible capabilities is complex, dynamic, and far from linear in its effects and consequences. This creates cataclysmic challenges for developing countries, particularly Africa, in all key economic sectors. The opportunities of AI, although achievable, stand at the cost of fiscal resources, skill-force, knowledge capacity, ICT infrastructure, governance and policy (Afful-Dadzie & Afful-Dadzie, 2017; Lalmuanawma, Hussain, & Chhakchhuak, 2020). These topical areas must be looked at, enforcing a deep understanding of the challenges to present tangible approaches to solve such idiosyncratic issues. In understanding this issue, the nature of African countries comes to bear.

In terms of ICT infrastructure, most African countries are lacking. Amidst this, both human and technical capabilities are lacking. As highlighted earlier in the digital divide, these challenges are proportional to the socio-economic development of most African countries. Apart from these issues, culture and literacy play a role. Certain cultures are quite difficult in adapting to new technologies, which affects the cognitive decisions of leaders and executives of industries in embracing the change (Tzachor, 2021). In addition, the curriculum in tertiary institutions that may have to be changed to suit the AI era requires strong human resources, which may be equipped knowledge-wise and hands-on-wise to impact the practice into students. This may also lack and challenging to achieve.

3.3 AI IN EDUCATION: THE PROBLEM

The sections earlier have elaborated on challenges besetting the educational sector in most developing countries that extend to implementing AI. Broadly, these challenges are classified into a human, economic challenge and institutional capacity. Fig. 3.1 illustrates a

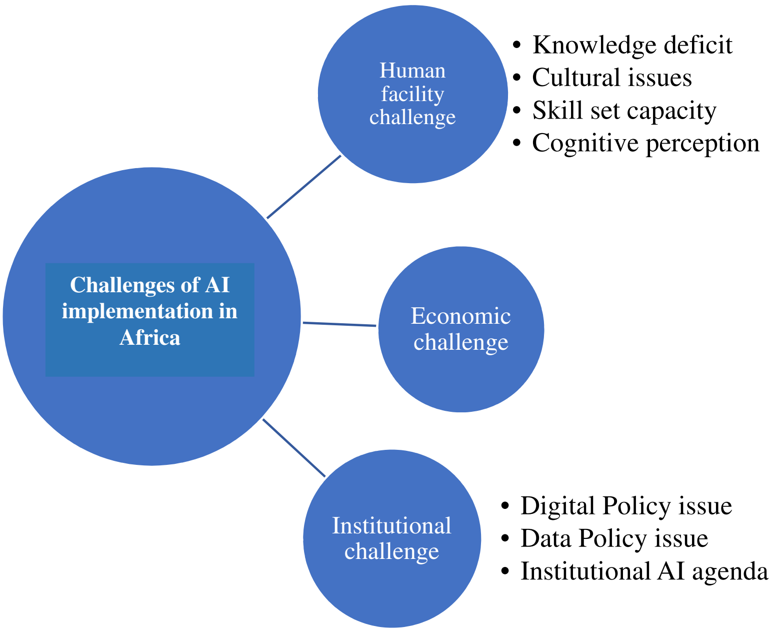

Fig. 3.1. Challenges of AI Implementation in Education in Africa.
Source: Author's own creation.

diagrammatic view of the challenges followed by a detailed discussion in the following section.

3.3.1 Human Facility Concerns

AI refers to non-human machines that have been programmed to have human-like intelligence and can thus perform tasks that humans can. Humans play a crucial role in the AI chain. Therefore, the engagement of humans is critical to AI's success. Here, the human could be the AI artefact's technical enabler, the recipient, the mediator or the moderator. The human persona must be engaged for AI adoption, especially in Africa. This can be discussed further from culture, literacy (education), skill and cognition.

3.3.1.1 Cultural Challenge

Tzachor (2021) defines culture as to how people conduct themselves or do certain things. Culture plays a major role in the adoption and implementation of an artefact. Individuals who are open to reinventions are creative. Extant research has explored how culture plays a role in technology adoption, especially how people perceive technology. For instance, Lee, Trimi, and Kim (2013) submitted that culture is individualistic and turns to allow members to seek information. This, to a greater extent, speeds up innovation, as there is varied thinking that promotes creativity. Other cultures depend on collective reasoning or others gaining access to specific information and depending on them to share (Vasudeva & Mogaji, 2021). This culture is high on imitation; thus, adoption depends on knowledge transfer and imitation. It is important to understand that this culture is beneficial for most countries as there is a literacy divide. Typically, changing or developing culture takes significant effort.

Moreover, organisations that do not integrate culture into their activities innovate. This slows down employee creativity, giving rise to information asymmetry (Abdulquadri, Mogaji, Kieu, & Nguyen, 2021; Back, 1993). This is a challenge for the propagation of AI in most developing countries. Yet, the developed economies that gave birth to culture in their business processes benefit from AI capabilities.

3.3.1.2 Knowledge Deficit

Advancement of knowledge is critical to the development of every country. Many countries have advanced based on knowledge acquisition and transfer. Knowledge management is critical in the technology adoption process. Studies have offered why knowledge is a critical component needed to overcome the technology adoption barrier. Given the developing nature of AI in most developing countries, understanding AI's nuances, theoretical implications and practical aspects are critical to its success in the educational environment (Mohamedbhai, 2011; Ndofirepi, Farinloye, & Mogaji, 2020). However, the dependent literacy rate in the African regions has become a bane to the proliferation of AI adoption (Mushemeza, 2016; Teferra, 2008). AI, as a domain, involves many complex moving parts of which mathematical science is a core. Applying the concepts to real-world scenarios is based on understanding what AI is. However, this is not so in many African regions due to the limited understanding of the phenomenon (Arakpogun, Elsahn, Olan, & Elsahn, 2021). By extension, the low understanding of the concepts of AI affects its applicability in real-life problem-solving. Hence, new knowledge, such as AI in education, is likely to meet resistance irrespective of the benefits. For instance, one researcher in this chapter experiences a similar situation in his academic institution where some academic staff have resisted the new advances of information systems, including AI and Machine learning.

3.3.1.3 Skill Set Capacity

As knowledge becomes scarce in Africa, so does the skill set. The developing nature of AI makes it a relatively complex activity. This requires the skill set necessary to undertake AI activities such as data mining and analysis. Predictive algorithms are embedded in AI machines to predict future variables and solve mundane tasks. This, however, demands technical skills to accomplish those tasks.

Moreover, these skills can only be complemented by adequate infrastructural resources and governmental push. Unfortunately, few governments have IT strategies embedded in their policies (Ceci, Masini, & Prencipe, 2019; Nguyen & Mogaji, 2022; Shiohira, Keevy, & Gibbs, 2018) to advance the course of AI development to generate users' buy-in into these new technologies. This essentially is an issue of misconception and a lack of clear technical understanding and application of AI (Gamji, Kara, Nasidi, & Abdul, 2021; Mohamedbhai, 2011). Moreover, given the limited skill set in Africa on AI, brain drain has taken centre stage where relatively few skilled persons are migrating to the advanced countries in search of greener pastures. This brings to fore infrastructural challenges besetting African countries in their AI implementation, particularly in its educational curricula. Therefore, it has become important to reform the educational sector according to the state-of-the-art technological paradigm to subvert this challenge.

3.3.1.4 Cognitive Perception of AI

The perception of technology elusively renders many families jobless, which increases the unemployment rate. This has dire consequences for adopting and continuous use of AI in Africa. Research in the 4IR shows that most industries will be AI-powered in a few years, especially on the production side (Kufakwababa, 2021). Regrettably, AI is threatening the existence of knowledge-based workers, bringing to bear a socio-economic challenge (Kufakwababa, 2021; Roy & Srivastava, 2021). This perception is also a reality that may arise in the not-too-distant future. This affects the perceptual acceptance of new technologies in regions where unemployment is already a major problem. Moreover, there is internal cognitive resistance to adopting technologies with such transformative impacts. This is an area worth exploring to understand and present tangible solutions.

Governments may embrace this view about the technology and, thus, affects their readiness to accept such technologies. Apart from this, another perception of AI or technologies is the 'armageddon theories' or Apocalyptic AI (Barney & Fisher, 2017; Geraci, 2008), which entails conspiracy theories or theories that indicate AI inception may bring the end of humans. These perceptions may be strong against the AI uptake in Africa. Sometimes, such ideologies are institutions of both religious populism and secular innuendos, and this can hinder the growth or adoption rate of AI.

3.3.2 Economic Challenge

AI and its related infrastructure comes with a huge cost to adopters and implementers. Beyond the technical and infrastructural challenges, prevailing economic circumstance has become a bane to successful AI deployment in Africa. Given the precarious economic conditions of most African countries, adopting and implementing AI is unrealistic. This economic challenge leads to low infrastructural growth and difficulty in building capacity. In jurisdictions where AI has been deployed, financial resource ready to sustain the digital infrastructure has become unsustainable. With a high debt to GDP ratio (Garba & Bellingham, 2021), the provision of basic educational resources is challenging to African governments. Given the demand for resource allocation in critical sectors of Africa's economy, issues of AI and its capabilities are often secondary to warrant urgent financial allocations. This also stems from the limited understanding of AI capabilities, given the limited attention in terms of capital expenditure that the area receives.

Technological services are best delivered through robust infrastructure. This infrastructure involves networks, data centres (databases) and operating systems. Several research papers that discuss technological adoption challenges in Africa show a lack of strong internet infrastructure. Internet connectivity is a major challenge in most African countries. Although there is a rise in telecommunication services, access is still a challenge, particularly in the continent's marginalised parts (Gastrow, 2020). Beyond access, the cost seems to be yet another constraint. The cost of the internet in Africa will be high compared to most developed economies. Existing network infrastructure is expensive, which has a corresponding influence on data charges. Despite the challenges associated with users, telecommunication service providers cannot expand AI-powered network infrastructure due to taxes imposed by governments on their services. For instance, despite the taxes imposed by the telecos, the Government of Ghana has presented a budget that seeks to impose a 1.75% electronic transaction on the citizens. This and many others block digital inclusion. The 4IR highly depends on the internet (Gastrow, 2020), and for Africa to garner the benefits, expensive data charges would not suffice as schools may have to pay high prices or charges of unreliable networks. These pose extended economic challenges such as high production costs, among others. The cost of using a network or internet bandwidth also affect learning which, in effect, affect citizens who may try to advance their knowledge in AI. Moreover, institutions that seek to integrate AI into their curricula may be retrained due to higher cost of Internet bandwidth.

3.3.3 Institutional Capacity

Institution, in this section, is referred to as the government capacity to initiate or propagate the AI agenda. Regardless of the challenges mentioned earlier, the major standpoint is the institutional capacity (Arakpogun et al., 2021; Hamdan, Hassanien, Razzaque, & Alareeni, 2021). Given the beneficial transformative power of AI, governmental bodies in Africa must create an enabling environment for AI to thrive. This enabling environment highly depends on the government's capacity in terms of digital policy strategies, data curation strategies and an institutional agenda towards AI. This means that governments may have to define heuristics that can guide how AI can be distilled into public sectors and businesses in the country. This can therefore strengthen the AI agenda.

3.3.3.1 Digital Policy Strategies Issue

The Organisation for Economic Co-operation and Development (OECD) motivates governments to develop digital strategies to utilise technologies (Ubaldi, 2020) effectively. New technologies often bring other associated risks and ethical issues that require an effective strategy. AI is a data-centric technology and an amalgam of other technologies; thus, it seems to depart from the linear traditional-based technologies. Therefore, digital policies must understand the power of AI and its effectiveness. However, many African countries have almost no digital policy strategy. If such policies exist, implementation becomes a challenge. More critical is the lack of experts in AI who understand the nature of AI and frame policy guidelines fit for their economic settings. Often, this results in implementing AI activities that do not meet the citizenry's demands. A lack of a clear digital policy strategy affects implementation and use.

3.3.3.2 Data Curation Policy Issue

As AI is a data-driven technology, data are paramount. However, one of the major challenges that affect the uptake of AI is the availability of properly generated data. Most public data are found in disintegrated systems, which inadvertently make data curation difficult. Getting readily available data in African countries is quite cumbersome, as agreed by (Ceci et al., 2019). Yet, regarding the continent's population and vibrancy of youth, coupled with the aggressive interest in internet activities, open data would offer a stronger push for the technology under discourse. However, it must be noted that there have been some data curation centres on the continent, such as Data Science Africa, MainOne and others (Arakpogun et al., 2021). However, these are not enough, and data curation strategies must be a governmental agenda rather than a private one. The government's inability to create such an enabling environment for data curation is a challenge to AI adoption. Many corporate bodies that may desire to enter the continent focusing on AI may find it difficult and thus resort to using data from other countries.

3.3.3.3 The Institutional AI Agenda

An agenda that can propel this transformative technology is lacking (Gastrow, 2020; Ubaldi, 2020). Some countries may have a digital agenda that they may purport to fit all technologies. However, AI's nature must be keenly taken cognizant as it presents new facets that affect work, education and lifestyle. While institutions in Africa are confronted with infrastructural and fiscal constraints to adequately leverage AI, the technology has a wide range of uses in businesses, including streamlining job processes and aggregating business data. The challenge presented here is linked to the propagation of AI among the various government sectors. Due to the lack of a clear-cut agenda, most public sectors are not utilising the capabilities of AI in their business process. This results in weak processes or bottlenecks in their processes, thus laybacking government to old forms of business. However, 4IR presents unique opportunities for countries, especially those in Africa (Ślusarczyk, 2018), to wean these benefits by setting an agenda for AI uptake.

3.4 AI IN EDUCATION: THE SOLUTION

Despite the challenges associated with AI implementation in Africa, the section discusses some good opportunities in the education sector. These opportunities are tilted towards implementing AI in education.

3.4.1 Rethinking AI Policy for Education

Undoubtedly, AI originated from academia since the term was coined from a workshop at Dartmouth in 1956, suggesting that the root of AI is deep-rooted in academia. Decades after its birth, educational institutions and regulatory bodies need to rethink their AI policies. There are deepening misconceptions about integrating AI in the educational curricula of developing countries despite its proven capabilities (Long & Magerko, 2020). Nonetheless, these concerns about AI's potential or speculative industrial uses must neither sway the discourse nor overshadow the impact that AI will have on nearly every policy domain.

Typically, AI advancement has revolved around the educational sector in developed economies. Furthermore, education is crucial to prepare future workforces for AI. However, the adoption of increasingly powerful technologies to facilitate learning isn't enough to close the AI skills gap (Chan & Zary, 2019). It also means going beyond rethinking the content and methods used to deliver instruction at all levels of education, hence the need to rethink AI policy in education. First, a deliberate attempt must be made to advance the study of science, technology, engineering and mathematics in all spheres of academia. This will set the ground for producing graduates with a strong analytical background to push the AI agenda. Second, a conscious attempt must foster collaboration among educational institutions and industry. Essentially, existing regulations on AI use in education must reflect a carefully calibrated and tailored guideline that prioritises data mining, big data analytics, machine learning (Završnik, 2019). Therefore, the need to create policies that foster an AI ecosystem that supports sustainable development and public policies at the international and national levels must be eagerly pursued. Finally, government support is the core for the advancement and sustenance of AI in education. This is imperative since governments, in most economic jurisdictions, are the regulators of educational institutions providing policy direction and financial and human resource availability (Jobin, Ienca, & Vayena, 2019; Kiraka, Maringe, Kanyutu, & Mogaji, 2021; Pedro, Subosa, Rivas, & Valverde, 2019). However, it must be acknowledged that while data collection and analytics are becoming increasingly powerful, their costs may be prohibitively high, especially for developing countries. Therefore, the costs of such data systems and technologies must be weighed and weighed against the benefits that may be realised.

Third, reordering teaching and learning experiences is required. Given that existing teaching and learning processes may be insufficient in capturing the nitty-gritty of AI, a substantial reordering of the dynamics of learning and learning to process enabled by AI is needed. Consequently, it will call for a total overhaul of the teaching requirements for which the format and later recommendations sought to address (Yigitcanlar, Desouza, Butler, & Roozkhosh, 2020) – fourthly, reanimating the teaching environment. Classroom settings in most under-developing countries are beset with poor teaching aids. Yet, teaching AI requires more than a physical space to include distributed, and network systems, cloud and virtual reality augmented tools. Ultimately, the impact will be unimaginable with students being exposed to state-of-the-art digital educational tools enabled by AI. A UNESCO report on AI in education for sustainable development submits that equity and inclusivity are a bane to AI development in developing economies (UNESCO, 2019). The report adds that, with AI advancement, the least developed countries are at risk of experiencing new technological, economic and social divides.

Another factor worth considering in AI educational policy is data privacy and security (Pedro et al., 2019). Data privacy and security are almost always brought up in discussions about data ethics. The main challenge is ensuring that personally identifiable information and individual privacy preferences are protected while using personal data (Osoba & Welser, 2017). It is also crucial to put safeguards to prevent data theft. This is made more difficult in education because young learners legally cannot provide express consent to collect and use their personal data (Manikonda, Deotale, & Kambhampati, 2018). While most developed countries are constantly putting in place data protection laws (Gravett, 2020), low-middle-income countries seem at alms length with comprehensive data protection laws (Netshirando, Munyoka, & Kadyamatimba, 2021). In the latter's case, the existing data privacy policies do not sufficiently protect data subjects (Carmody, Shringarpure, & Van de Venter, 2021).

3.4.2 Rethinking the AI Curricula

The heightening demand for adaptive and collaborative learning methodologies calls for a rethink of AI curricula, especially in the Global South, where AI adoption and implementation in education is in its infancy. Instances where 'machine teachers', also known as AI teaching assistants, have been created in most developed economies, such as the United States, in response to the growing demand for intelligent teaching and learning (Kim, Merrill, Xu, & Sellnow, 2020), it set the pace for a rethink of AI curricula in Africa. Deep networks, such as humans, have learned more effectively when samples are organised and presented logically in the educational curriculum. Rethinking and reworking the educational curriculum to prepare learners for the increasing presence of AI in all aspects of human activity cannot be understated. First, a new curriculum for a digital and AI-powered world must expand on the importance of teachers and students improving their digital competency frameworks. While some current initiatives such as the Global Framework to Measure Digital Literacy and the ICT Competencies and Standards from the Pedagogical perspective (UNESCO, 2019) exist, they could not add substantial value to the AI dimension globally in educational curricula. The demand now is to broaden the discussion of the curricular dimension to include new experiences for developing computational thinking in schools. Africans should strive to contextualise their own AI curriculum in all situations rather than always waiting to adapt from the West.

Second, a pragmatic approach with enhanced AI capabilities through advanced education and training is needed. However, the question of what each country can due to prepare for an AI-powered world is still underexplored. Therefore, developing comprehensive plans to address this issue is important. Existing AI curricula fall under two main categories, according to (Kersting, 2020): first-generation AI systems follow well-defined rules written by programmers to cover every eventuality. Second-generation AI systems employ statistical learning to solve a specific problem.

Nonetheless, the recent trend in image classification, which is an embodiment of AI, has pushed for a rethink of AI curricula. Hence, a third force is needed to project AI beyond programmable algorithmic tools. Here, an AI tool can learn to communicate and reason like humans and recognise and adapt to new situations (Kaplan & Haenlein, 2019). In most circumstances, AI tools used in developing countries form part of the second wave where they act as specialised systems proficient in specific and well-defined tasks. However, they are frequently not robust, and without comprehensive retraining, they frequently fail in even minimally different situations. For example, many second-wave AIs for visual scene understanding is beguiled by an object in a non-canonical orientation and context (Ursyn, 2018, pp. 164–215). Therefore, as Kersting (2020) observed, there is a need to bring together another view of combined low-level perception and high-level reasoning in new AI curricula. By this, individuals are better positioned to capture, understand and utilise AI.

Again, some streams of studies (e.g. Sperrle et al., 2021; Van Zoelen, Van Den Bosch, & Neerincx, 2021) have suggested that AI should be co-adaptive by modelling the dynamic behaviour of human–computer behaviour to adapt effectively to machine learning. Van Zoelen et al. (2021) also suggest balancing tasks that automate and tasks we humans can still contribute. Additionally, incorporating computational models for human intelligence into AI systems allows machines to learn as much about the world as humans do simultaneously and with the same flexibility. Meeting these demands not only pushes AI but also pushes humans by offering a once-in-a-lifetime opportunity to rethink fundamental AI problems and methods, ranging from hardware and software design to databases and robotics to human–computer interaction and software engineering (Kersting, 2020). Finally, reaching out to another field of endeavour, such as cognitive and management sciences, will offer a broader view of combining reasoning and learning (Vogan, Alnajjar, Gochoo, & Khalid, 2020).

3.4.3 Rethinking of the AI Artefact Creation Methodology

Strengthening synergy between stakeholders in the educational ecosystem has been recommended in promoting innovation towards improved corporate performance. According to Merz, Zarantonello, and Grappi (2018), co-creation is the process by which stakeholders and organisations collaborate to create value from their products or services and brands. Typically, co-creating an AI artefact with stakeholders does not imply institutions are interested in pleasing their customers. Instead, the objective of co-creation is to collaborate and co-create the experiences of each other through dialogue (Hinson & Mogaji, 2020; Prahalad & Ramaswamy, 2004). Active dialogue is significant in co-creating an AI artefact since it generates interactivity and mutual learning among stakeholders to generate the desired product. Note that collaborating to generate an AI artefact design brings about a more advanced and acceptable product. Prior studies specifically (Loureiro, Romero, & Bilro, 2020) aver that co-creation, an embodiment of process creation interactive system environment, is symbiotic to open innovations. Loureiro et al. (2020) further report that interest in investigating co-creation among researchers has heightened since 2010 when firms and institutions are increasingly engaging stakeholders in their business process improvements. AI, which is still developing in most developing countries, needs to embrace effective stakeholder engagements in its product design.

Moreover, given the three main characteristics of the AI learning process (supervised, unsupervised and reinforcement learning), several skill set issues, that characterise humans as out of reach of AI, persist (Kaplan & Haenlein, 2019). Yet, tremendous progress has been made to close the gap between AI and human capabilities over the years. Even though the adventure seems unlikely, the probable solution is a collaborative effort between stakeholders in creating adaptive AI products and services. Hevner, March, Park, and Ram (2004) advocated for the co-creation of an artefact through design science research. Hevner et al. (2004) highlighted rigorous evaluation of an artefact, with end-users, as a central focus of design. The Hevner et al. (2004) cycle encompasses three components: The environment, design and knowledge base. The environment makes up the application domain's understanding, including the people, organisation and technical systems. The goal is to plan objectives for developing solutions to design and evaluation. The design stage involves the actual development of the artefact. However, the design stage undergoes a vertical stepwise process (Design cycle: build and evaluate). In all the processes, stakeholders, especially those in education, are expected to co-create a solution to the real-world problem. The environment and design stages iterate until a desirable outcome is attained, which makes up the relevance cycle. The artefact is expected to add knowledge to the domain for which the study was undertaken. Thus, contributing to the knowledge base largely relies on the developed artefact. The iterative process at this stage is the rigour cycle.

Arguably, AI has redefined the concept and meaning of work in the 4IR by transforming how businesses are scaling up their operations through expediency, flexibility, personalisation and enhanced decision-making (Dwivedi et al., 2021; Feldman, 2017; Woo, 2020). However, the fear of job losses and the dislocation of employees from AI adoption seem to be the subject of worry even among skilled workers because organisations get better margins when jobs are automated. This harm to humankind also heightened the need to rethink the AI design process (Song et al., 2021). Note that co-creation between humans and AI is based on enhancing each other's strengths such as leadership, cohesion, originality, social intelligence, scalability and quantitative abilities (Feldman, 2017). Therefore, every stakeholder in the co-creation process has a role in designing algorithms for AI artefacts (Mramba, Apiola, Kolog, & Sutinen, 2016; Wu et al., 2021).

3.4.4 AI-Based Collaborative Learning Platforms

Collaborative learning in the Global South has peaked in the wake of the COVID-19 (Daniel, 2020). Collaborative learning platforms assist a group of two or more people in achieving a common goal or objective by sharing educational content or resources. For instance, videoconferencing and instant messaging teleconferencing have been among the typified collaborative platforms for learning in recent times. AI has elevated collaborative learning through intelligent agents, learner support chatbots, matching of a learner to learner/teacher while placing learners in full control of their learning activities. AI-based collaborative learning platforms offer easy access to lifelong learning companionship through intelligent tutoring systems (McLaren, Scheuer, & Mikšátko, 2010; Oginni, Mogaji, & Nguyen, 2022). With the proliferation of social media and smart mobile applications, learners expect an expedited response to their queries during their learning process, thus making the capabilities of AI-based collaborative learning a compendium of knowledge management (Hamza-Lup & Goldbach, 2019).

Given the proliferation of smartphones and social media usage among students in most developing economies (Farinloye, Mogaji, & Kuika Watat, 2021; Olaleye, Ukpabi, & Mogaji, 2020), students are better positioned to optimise their learning. Studies have shown that AI-enabled collaborative tutoring systems stand to increase students' performance (Echeverria et al., 2020; Hamza-Lup & Goldbach, 2019; McLaren et al., 2010; Yang et al., 2021). It is essential to state that AI-enabled collaborative learning tools incorporate machine learning algorithms, modelled with several learning requirements fit for curriculum activities (Hamza-Lup & Goldbach, 2019). Learning can also be facilitated by observing the dialogue and reflecting on the dialogue by submitting questions and answers, and interacting between the face-to-face and online environments. Echeverria et al. (2020) emphasised that a hybrid controlled intelligent collaborative system is needed to advance the transition of students learning and make it more adaptive for different classroom settings.

3.4.5 Tracking Learners' Opinions With NLP Techniques

Succinctly, educational institutions across the globe are increasingly looking for innovative ways to support learners and make learning activities more engaging (Zawacki-Richter, Marín, Bond, & Gouverneur, 2019). For instance, tracking students' progress in a traditional classroom setting is challenging. For this reason, convergent tools are proffered to aid in tracking students' learning progress in real-time. This is essential in the era where learning and teaching have been migrated online. Beyond the traditional student feedback forms, educational institutions worldwide are increasingly interested in using different innovative approaches to gauge student attitudes towards their learning experience (Sharma et al., 2019; Kolog, Montero, & Sutinen, 2016). The renaissance of AI has helped educators discover students' needs for real-time interaction in developed countries. This provides a comprehensive approach to tracking and measuring teaching and learning activities. Pham et al. (2020) maintain that opinion mining is one means of tracking students' opinions. This technique uses natural language processing (NLP) and machine learning techniques to explore students' emotional and sentimental experiences. The data sources for analysis are often from students' feedback and interaction during ongoing teaching and learning. For example, Kolog et al. (2018) developed an e-counselling system that tracks students' sentiments and opinions about the academic trajectory.

Research has shown that learners are the centre of the educational system, and their feedback can impact instructors when they get honest responses from their students (Fyfe & Brown, 2020; Hays, Kornell, & Bjork, 2010; Mag, 2019). In their recent study, Fyfe and Brown (2020) sought to identify how a teacher's explicitly stated expectations influence students' ability to learn from corrective feedback during problem-solving. The study suggested that students' opinions become beneficial when instructors help learners set their success goals. As a matter of interest, educational researchers are often interested in understanding students' learning experiences and attitudes towards learning activities through feedback (Mag, 2019). This makes opinion mining of student learning experiences and attitudes crucial. Yet, Pham et al. (2020) aver that given the value of students' opinions, the input can only be clear at the end of the traditional learning activity. This implies students' emotions can be observed during the learning process. Where AI analytics is applied, particularly in online learning, opinions via posts can be processed in real-time for instructors to make informed decisions (Kim et al., 2020). Therefore, we propose that schools in African countries integrate a system to analyse and provide feedback on students' learning experiences and opinions.

3.5 CONCEPTUAL FRAMEWORK

In Fig. 3.2 (Inspired by Hevner et al., 2004), we present a conceptual framework of the implementation of AI in the education sector. The figure was adduced by synthesising the discussions aforementioned on the problems and proposed solutions of AI implementation in Africa. The framework is linearly composed of three main components: the problem domain, the proposed solutions to AI implementation and the outcome. As earlier established in this chapter, AI implementation in the education sector is bedevilled with the challenges that emanate from society, society in education and the AI domain. It, therefore, includes the people (including stakeholders), educational institutions and the existing technologies. The people are the education stakeholders who drive the implementation of AI in education. The second component is the proposed solutions to AI implementation in the education sector. As suggested in this study, the proposed solutions are first to revamp/develop AI policies to guide the use and implementation of AI in the education sector, particularly in digital and data policies.

The existing educational curriculum needs to be restructured to include practically oriented courses in AI, solving the AI knowledge deficit. Furthermore, tools such as AI-enabled collaborative learning platforms should be encouraged in the education sector to promote more effective remote learning. The COVID-19 has awakened the need to promote remote learning. This brings about the need to enhance collaborative and online learning by mimicking the face-to-face classroom setup. Therefore, our considered view is

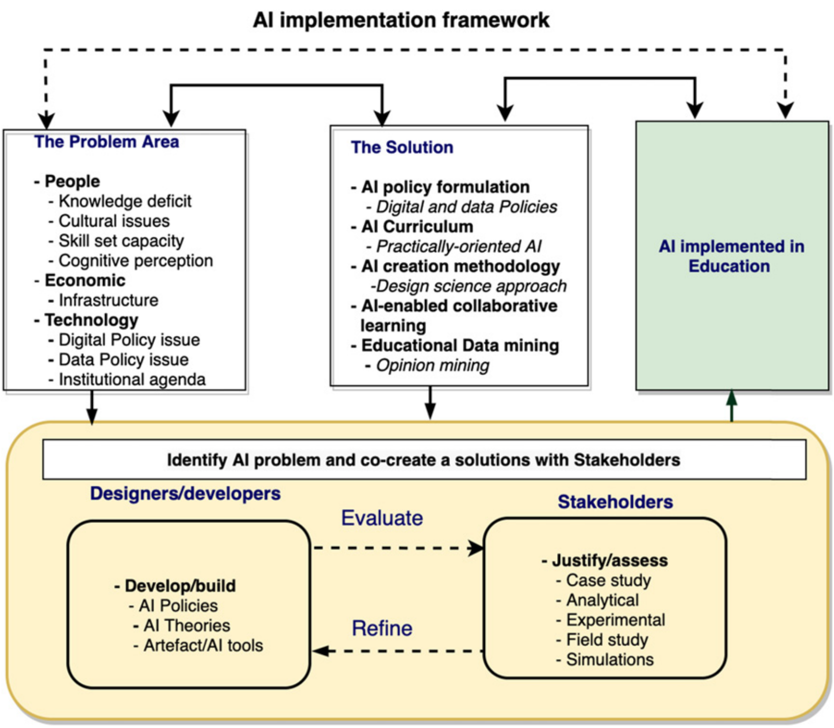

Fig. 3.2. Conceptual Framework for AI Implementation in Africa.
Source: Author's own creation.

that NLP resources could be leveraged to process natural language from students' data collected via online and remote learning. The objectives of the problem domain and the proposed solutions should be explored and co-created with education stakeholders. The co-creation process should be driven by agility, where stakeholders are made to provide feedback to the solution that aims to solve societal problems within the education sector, as Fig. 3.2 suggests. Thus, end-users and other stakeholders will be motivated to use AI solutions given their involvement in the co-creation process.

3.6 IMPLICATIONS FOR THEORY AND PRACTICE

The findings of this study have far-reaching implications for theory and practice. As a key contribution to literature, the proposed conceptual framework integrates AI implementation competing forces while proffering mitigating solutions. Drawing on the conceptual framework (Fig. 3.2) for the implementation of AI in Africa, this chapter offered a practical solution to the myriads of challenges besetting the implementation of AI for sustainable education in Africa. Given the lacklustre approach to AI implementation in Africa, our extensive review, as presented in this chapter, has successfully exhumed the criticalities of the challenges faced by AI implementation in Africa. This has pioneered a rethink of AI implementation among researchers in Africa. While research on AI has been sparsely discussed in Africa, particularly in education, our proposed framework will invigorate the existing inertia and supplement further empirical studies in AI focusing on the developing economies context. Again, the conceptual framework seeks to guide researchers on the need to water down the countervailing forces of AI implementation in Africa. The implementation strategy proposed in this chapter is important given the developing nature of AI.

Furthermore, the phenomenon is still in its infancy, as observed in the review relative to developing economies. Thus, this study sets the stage for further empirical analysis to leverage the identified strategy to unearth the optimal solutions to mitigate AI implementation challenges. Moreover, the framework will serve as a springboard for researchers to expand the scope by leveraging the identified constructs in the framework.

Given the continuous examination of Africa's educational curricula, the study unravelled the barriers to implementing AI in Africa. Based on the identified barriers, solutions are proposed to implement AI in Africa. Thus, stakeholders of education, particularly governments and policymakers, could leverage these findings to develop actionable policies for AI implementation in Africa. The affordances of AI have been established in literature with little light on the implementation challenges, particularly among African countries; this puts this chapter in good light. This chapter has successfully provided challenges of AI which have threatened its smooth implementation in Africa. Thus, offering managers of institutions the needed understanding and strategies to eradicate the deepening challenges of AI through the co-creation of AI curricula. As suggested, the need to restructure existing AI curricula to include practically oriented courses of not far-fetched, giving the need to embrace innovative learning processes leveraging state-of-the-art tools. Yet, the resilience of AI deployment in Africa has been questioned for its vulnerability, subverting the cause of effective implementation. Thus, this study's implementation framework presents a robust guide to managers of AI in both private and public institutions across the African continent.

3.7 CONCLUSION

AI has existed since the early 1950. Machine learning, deep learning and NLP are all products of AI advancement. These technologies, if fully harnessed, have the potential to improve flexibilities in human activities, including the educational sector. While the developed economies have explored and advanced research in AI technologies, Africa lags in its implementation. Implementing AI could be attributed to several challenges, as enumerated in this chapter. We classified the challenges according to human, economic and institutional challenges. Following, we have proposed possible solutions to curtail the implementation issues of AI in education. The study found a knowledge deficit, policy direction and practically oriented AI curriculum as a major hindrance to AI implementation in Africa. This finding is a wake-up call for academic institutions, especially higher education institutions, to rethink the implementation of AI in Africa. We recommend the strengthening of academic-industry partnerships towards promoting AI education in Africa. It is, therefore, our considered view that experts from the computing domain, especially those who would be made to participate in the implementation process of AI, promote the 4IR in Africa.

This study's challenges and proposed solutions for AI implementation in Africa are limited to only the education sector. Based on this study's findings, we suggest future studies to focus on the dynamic capabilities of the country's education sector, thereby establishing the readiness for AI implementation in Africa. In addition, we recommend further studies to explore the impact of AI implementation on students' academic performance. Finally, we envision that the proffered solutions, if implemented, could mitigate the deepening misconceptions and challenges bedevilling AI implementation in Africa.

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